

# Food Delivery Time Prediction Using Linear Regression

**Dataset:** Delivery\_food.csv

**Model to be Used:** Linear Regression Only

**Total Questions:** 30

- Section A: Linear Regression (20 Questions)
- Section B: Visualization (10 Questions)

Students must use Python (Pandas, NumPy, Matplotlib, Scikit-Learn).

## SECTION A: LINEAR REGRESSION (20 QUESTIONS)

**Q1.** Load the dataset and display the first five rows. Describe the purpose of each column.

**Q2.** Find the total number of records and features in the dataset.

**Q3.** Identify:

- (a) Numerical variables
- (b) Categorical variables
- (c) Target variable

**Q4.** Check for missing values in all columns. Identify which column contains missing data.

**Q5.** Handle the missing value in the Time\_of\_Day column using an appropriate method and justify your choice.

**Q6.** Remove the Order\_ID column and explain why it should not be used in model training.

**Q7.** Apply One-Hot Encoding to the following categorical columns:

- Weather
- Traffic\_Level
- Time\_of\_Day
- Vehicle\_Type

**Q8.** Explain why categorical features must be converted into numerical form before applying Linear Regression.

**Q9.** After encoding, report the new total number of input features.

**Q10.** Define the feature matrix  $X$  and target vector  $y$ .

**Q11.** Train a Linear Regression model using the dataset.

**Q12.** Write the regression equation using:

- Intercept
- Coefficients of all features

**Q13.** Predict the delivery time for all input records using the trained model.

**Q14.** Add a new column called Predicted\_Delivery\_Time to the dataset.

**Q15.** Compute the residuals (Actual – Predicted Delivery Time) for all observations.

**Q16.** Calculate the Mean Absolute Error (MAE) of the model.

**Q17.** Calculate the Mean Squared Error (MSE) of the model.

**Q18.** Calculate the  $R^2$  Score and explain its meaning in the context of this dataset.

**Q19.** Identify the top three features that have the strongest impact on Delivery Time based on model coefficients.

**Q20.** Based on your results, provide three practical business recommendations to reduce food delivery time.

## Use **Matplotlib only** for plotting.

- Q21.** Plot a histogram of Distance\_km. Describe the distribution.
- Q22.** Plot a histogram of Preparation\_Time\_min. State whether the data is uniformly distributed.
- Q23.** Create a bar chart showing average Delivery\_Time\_min for each Traffic\_Level.
- Q24.** Create a bar chart showing average Delivery\_Time\_min for each Weather condition.
- Q25.** Create a bar chart showing average Delivery\_Time\_min for each Vehicle\_Type.
- Q26.** Draw a scatter plot of Distance\_km versus Delivery\_Time\_min. Comment on the trend.
- Q27.** Draw a scatter plot of Courier\_Experience\_yrs versus Delivery\_Time\_min. Explain the relationship.
- Q28.** Create a boxplot of Delivery\_Time\_min grouped by Time\_of\_Day.
- Q29.** Plot a scatter plot of **Actual vs Predicted Delivery Time**.
- Q30.** Plot the residuals against the predicted values and comment on model behavior.

## IMPORTANT INSTRUCTIONS TO STUDENTS

1. Use only Linear Regression.
2. Do not use advanced models such as Decision Trees, Random Forest, or XGBoost.
3. All plots must be properly labeled with titles, X-axis, and Y-axis.
4. Provide interpretation for every evaluation metric and visualization.